

MOUNT KENYA UNIVERSITY
SCHOOL OF EDUCATION
DEPARTMENT OF CURRICULUM AND INSTRUCTION

**TEACHERS' COMPETENCE AND ITS INFLUENCE ON THE EFFECTIVE
IMPLEMENTATION OF THE COMPETENCY-BASED CURRICULUM IN
PUBLIC PRIMARY SCHOOLS IN HABASWEIN SUB-COUNTY, WAJIR
COUNTY, KENYA**

[FULL NAME OF STUDENT]

[REGISTRATION NUMBER]

A Research Proposal Submitted to the School of Education in Partial Fulfilment of the
Requirements for the Award of the Degree of Master of Education (Curriculum
Studies) of Mount Kenya University

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DECLARATION

This research proposal is my original work and has not been presented for the award of a degree or for any other award in any other university.

Signature: _____ Date: _____

[Full Name of Student]

[Registration Number]

Supervisors' Approval

This research proposal has been submitted for examination with our approval as the university supervisors.

Signature: _____ Date: _____

[Name of Supervisor 1]

Department of Curriculum and Instruction, School of Education, Mount Kenya University

Signature: _____ Date: _____

[Name of Supervisor 2]

Department of Curriculum and Instruction, School of Education, Mount Kenya University

DEDICATION

I dedicate this work to my family, whose patience, prayers and encouragement have carried me through my postgraduate studies, and to the teachers of Habaswein Sub-County, who continue to serve their learners with dedication under some of the most demanding conditions in the country.

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ABBREVIATIONS AND ACRONYMS

APHRC	African Population and Health Research Center
ASAL	Arid and Semi-Arid Lands
BOM	Board of Management
CBA	Competency-Based Assessment
CBAM	Concerns-Based Adoption Model
CBC	Competency-Based Curriculum
CBE	Competency-Based Education
CIDP	County Integrated Development Plan
CPD	Continuous Professional Development
CSO	Curriculum Support Officer
CVI	Content Validity Index
ICT	Information and Communication Technology
KICD	Kenya Institute of Curriculum Development
KNEC	Kenya National Examinations Council
MKU	Mount Kenya University
MoE	Ministry of Education
NACOSTI	National Commission for Science, Technology and Innovation
PCIs	Pertinent and Contemporary Issues
PWPER	Presidential Working Party on Education Reform
SPSS	Statistical Package for the Social Sciences
TPACK	Technological Pedagogical Content Knowledge
TSC	Teachers Service Commission
UNESCO	United Nations Educational, Scientific and Cultural Organization

ABSTRACT

Kenya introduced the Competency-Based Curriculum (CBC) in 2017 to raise learners who can think critically, solve problems, communicate well and use what they learn in real life. Whether the reform succeeds rests largely on teachers. They are expected to understand what the curriculum is trying to achieve, to teach in learner-centred ways and to assess learners through a range of authentic tools. Studies from different parts of the country suggest that many teachers have not yet developed this competence, and the gap may be widest in marginalised arid and semi-arid counties such as Wajir. This study therefore sets out to assess teachers' competence and its influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County, Wajir County, Kenya. It has four objectives: to assess teachers' knowledge and understanding of the principles and practices of the CBC; to analyse their pedagogical competence in implementing it; to identify the factors that affect their competence; and to establish the relationship between teachers' competence and effective implementation. The study draws on the Concerns-Based Adoption Model, the Teacher Professional Competence Model and the Technological Pedagogical Content Knowledge framework. It will use a convergent parallel mixed methods design that combines a descriptive survey with a correlational approach. The target population is about 42 public primary schools, 42 head teachers, 336 teachers and 3 Curriculum Support Officers. Stratified random sampling will be used to select 21 schools, and a sample of 183 teachers, worked out with a finite population formula, will be spread proportionately across the strata. The head teachers of the 21 schools and all 3 Curriculum Support Officers will also take part. Data will be gathered through questionnaires, semi-structured interviews and classroom observation. The instruments will be piloted in Wajir South Sub-County, with content validity checked by experts and reliability tested using Cronbach's alpha. Quantitative data will be analysed in SPSS using descriptive statistics, Pearson correlation and multiple linear regression, while qualitative data will be analysed thematically. The findings should help the Ministry of Education, the Teachers Service Commission, the Kenya Institute of Curriculum Development and the County Government of Wajir to plan better-targeted support for teachers in marginalised areas.

Keywords: Teachers' competence, Competency-Based Curriculum, curriculum implementation, pedagogical competence, marginalised areas, Wajir County

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Schools in many parts of the world are rethinking what it means to learn. For a long time, a good pupil was one who could recall facts and reproduce them in an examination. Today, many education systems want more than that. They want learners who can use what they know to solve problems, work with others and make sound decisions in everyday life. This approach, usually called competency-based education, gives weight to skills, values and attitudes rather than to the amount of content a learner can remember (Jentsch & König, 2022). The change is happening at a difficult time. UNESCO and the International Task Force on Teachers for Education 2030 (2024) estimate that the world will need 44 million more primary and secondary teachers by 2030, about 15 million of them in sub-Saharan Africa. The same report warns that hiring alone will not close the gap. Teachers who are already in service also need help to build the kind of competence that newer curricula ask of them.

What, then, makes a teacher competent? A certificate is only the starting point. Jentsch and König (2022) see teacher competence as a blend of professional knowledge, practical skill, beliefs and motivation, which together allow a teacher to handle the everyday demands of a real classroom. Drawing on large international studies, they show that these qualities grow over time, during initial training and later through professional development, and that they shape the quality of teaching that learners receive. Mishra et al. (2023) make a related point: teachers' knowledge cannot stand still. As curricula and technologies change, teachers have to keep bringing together what they know about content, about teaching and about the tools available to them. Put simply, a curriculum is only as strong as the people who teach it.

Several African countries have introduced competence-based curricula over the past decade, and their experience is instructive. Wawire et al. (2026) reviewed studies from Ghana, Kenya, Rwanda, South Africa, Tanzania, Uganda and Zambia. On paper, these countries all promote learner-centred teaching and formative assessment. In many classrooms, though, lessons are still dominated by teacher talk, note copying and end-of-term tests. Kasuga and Kalolo (2025) describe similar tensions in Njombe Region, Tanzania, where poorly prepared teachers, an uneven spread of resources and parents' continued focus on examination results pulled against the new approach. Uganda tells much the same story. Teachers in Gulu City reported large classes, few materials and too little training (Kidega et al., 2024), and Barasa et al. (2025) worry that schools

serving poorer families are falling behind, so that a reform meant to reduce inequality may end up widening it.

Kenya started its own reform in 2017, when the Competency-Based Curriculum (CBC) began to replace the 8-4-4 system, beginning with pre-primary and the lower primary grades. The new curriculum aims to nurture every learner's potential through seven core competencies: communication and collaboration, critical thinking and problem solving, imagination and creativity, citizenship, digital literacy, learning to learn, and self-efficacy (Akala, 2021). Learners now spend two years in pre-primary, six in primary, three in junior school and three in senior school before moving on to tertiary education. In 2022, the government set up the Presidential Working Party on Education Reform (PWPER) to review how the reform was working. The Working Party kept the competency approach. It did, however, ask for fewer learning areas, a lighter assessment burden and a much stronger investment in teacher capacity (Presidential Working Party on Education Reform [PWPER], 2023). Since then, the reform has increasingly been described as Competency-Based Education (CBE), although what is expected of teachers in the classroom has hardly changed.

Assessment sits at the heart of the Kenyan reform. The Kenya National Examinations Council (KNEC, 2021) introduced a competency-based assessment framework in which teachers combine summative tests with a range of formative tools, among them observation schedules, rating scales, checklists, portfolios, projects and rubrics. The framework takes for granted that teachers can design authentic tasks, judge learners' work against clear criteria and keep accurate records for every child. That assumption does not always hold. Several years into the reform, Macheso, Kati, and Wafula (2024) found that many primary teachers in western Kenya still leaned heavily on written tests, while portfolios, anecdotal records and projects were used far less often. The competence the curriculum depends on, in other words, has not yet taken root among many of the teachers expected to deliver it.

Other Kenyan studies paint a similar picture. Research in Nairobi, Nyamira, Kirinyaga, Kinangop, Moiben and Bomet has found, again and again, that teachers were either not trained for the new curriculum or were trained in a rush that left gaps in their understanding (Isaboke et al., 2021; Karua et al., 2025; Keter & Wabuke, 2025; Mohamed et al., 2022; Mutai et al., 2024; Okeyo & Mokua, 2023). One finding stands out. Mwangi et al. (2026) report that 95% of the teachers they studied had attended CBC training, yet 65% still viewed the curriculum negatively. Sitting through a workshop, it seems, does not by itself make a teacher competent or committed. Mwang'ombe (2021) described teachers' grasp of the CBC as vague, a weakness that showed in their planning, their teaching and their assessment. In a classroom-based evaluation, the

African Population and Health Research Center (APHRC, 2022) found that some teachers were delivering effective CBC lessons, but many lessons fell short on learner engagement and on the integration of core competencies.

These problems are likely to be sharper in the arid and semi-arid lands (ASAL) of northern Kenya. Hussein (2023) found that poor infrastructure, scarce learning resources and limited training were holding back the CBC across the region. In Township Ward, Garissa County, more than 60% of public primary teachers had not received comprehensive in-service training on the curriculum (Ongubo, 2025). Wajir County, the setting of this study, faces many of the same pressures. Its County Integrated Development Plan lists a shortage of teachers, long distances to school, recurrent drought that disrupts enrolment and attendance, and a pastoralist way of life that makes continuous schooling hard to sustain (County Government of Wajir, 2023). Insecurity in the border areas has added to the strain, as many teachers from outside the region have sought transfers, leaving schools short-staffed and increasingly dependent on teachers hired by Boards of Management.

Habaswein Sub-County lies in the south of Wajir County, along the road between Garissa and Wajir. Most of its public primary schools serve pastoralist and agro-pastoralist families, and several run low-cost boarding facilities so that children from mobile households can stay in school. The teachers here work in tough conditions. Classes are often large, teaching materials are few, electricity and internet are unreliable and chances for professional development are rare. Even so, they teach the same curriculum as teachers in well-equipped urban schools, and their learners sit the same national assessments. It is worth asking, then, how well prepared these teachers are to deliver the CBC, what shapes their competence and how that competence shows up in their classrooms. These are the questions this study sets out to answer.

1.2 Statement of the Problem

For the CBC to work, teachers need to understand what it is trying to achieve. They need to teach in learner-centred ways, weave core competencies and pertinent and contemporary issues into their lessons and assess learners through authentic tasks. In an ideal situation, every public primary teacher in Kenya would have these abilities, and a child in Habaswein would enjoy much the same classroom experience as a child in Nairobi or Nakuru. The government has taken real steps in this direction through national training, the supply of curriculum designs and textbooks, and the work of Curriculum Support Officers (PWPER, 2023).

The reality falls short of this ideal. Research from across the country shows that many teachers have only a partial understanding of the principles of the CBC, struggle to

prepare schemes of work and lesson plans in the required format and use a narrow set of assessment tools (Karua et al., 2025; Macheso, Kati, & Wafula, 2024; Mwang'ombe, 2021). In northern Kenya, teacher shortages, insecurity, weak infrastructure and patchy in-service training make matters worse (Hussein, 2023; Ongubo, 2025). Where teachers lack the competence they need, implementation tends to stay on the surface. The vocabulary of the new curriculum is adopted, but lessons remain teacher-centred and driven by examinations (Awiti, 2023; Wawire et al., 2026). Learners in marginalised areas then miss out on the very skills the reform was designed to build, and old inequalities grow deeper (Barasa et al., 2025).

Plenty of studies have looked at teacher preparedness for the CBC, but most were carried out in urban or relatively well-served counties such as Nairobi, Machakos, Kirinyaga, Nyandarua and Uasin Gishu (Isaboke et al., 2021; Mogere, 2022; Mutua & Waweru, 2023). Wajir County has received little attention, and the researcher knows of no study that has examined teachers' competence and CBC implementation in Habaswein Sub-County. A second weakness is one of measurement. Many earlier studies judged preparedness mainly by whether teachers had attended training. They did not assess what teachers actually know, how they teach or how they assess learners, and they rarely tested statistically how these abilities relate to the quality of implementation. Without evidence of this kind, the Ministry of Education, the Teachers Service Commission and the County Government of Wajir have little to guide targeted support. This study responds to that gap by assessing teachers' competence and establishing its influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County, Wajir County.

1.3 Purpose of the Study

The purpose of this study is to assess teachers' competence and to establish how it influences the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County, Wajir County, Kenya. The aim is to produce evidence that can guide teacher support and education policy in marginalised settings.

1.4 Research Objectives

1.4.1 General Objective

The general objective of the study is to assess teachers' competence and its influence on the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County, Wajir County, Kenya.

1.4.2 Specific Objectives

The study will be guided by the following specific objectives:

- i. To assess the level of teachers' knowledge and understanding of the principles and practices of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- ii. To analyse the level of teachers' pedagogical competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- iii. To identify the factors affecting teachers' competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.
- iv. To establish the relationship between teachers' competence and the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County.

1.5 Research Questions

The study will seek to answer the following research questions:

- i. What is the level of teachers' knowledge and understanding of the principles and practices of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- ii. What is the level of teachers' pedagogical competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- iii. What factors affect teachers' competence in implementing the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?
- iv. What is the relationship between teachers' competence and the effective implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County?

1.6 Research Hypotheses

The fourth objective will be tested statistically through the following null hypotheses, each at the 0.05 level of significance:

- H01: There is no statistically significant relationship between teachers' knowledge and understanding of the CBC and the effective implementation of the CBC in public primary schools in Habaswein Sub-County.

H02: There is no statistically significant relationship between teachers' pedagogical competence and the effective implementation of the CBC in public primary schools in Habaswein Sub-County.

H03: Teachers' overall competence has no statistically significant influence on the effective implementation of the CBC in public primary schools in Habaswein Sub-County.

1.7 Justification of the Study

The CBC is one of the most ambitious education reforms Kenya has ever undertaken, and the first learners to go through it have only recently moved into junior and senior school. This is the right moment to ask how well it is working in different parts of the country. Equity is a second reason. The reform was meant to widen opportunity, yet experience elsewhere in the region shows that competence-based reforms can deepen the divide between advantaged and disadvantaged schools when support for teachers is uneven (Barasa et al., 2025; Kasuga & Kalolo, 2025). Habaswein is exactly the kind of place where such a divide would appear first.

There is also a practical case. The PWPER (2023) called for stronger teacher capacity building, and the Teachers Service Commission continues to spend on in-service training. That money will be better used if it is guided by evidence on where teachers are strong, where they struggle and what stands in their way. Finally, Kenyan research on the CBC has been concentrated in central, western and urban areas. A study set in the North Eastern region will add a missing voice to the national picture.

1.8 Significance of the Study

The findings should be of use to several groups. The Ministry of Education and the Kenya Institute of Curriculum Development (KICD) could draw on them to adjust curriculum support for ASAL counties, for example by simplifying materials or offering assessment guidance that suits local conditions. The Teachers Service Commission (TSC) could use them to shape in-service training and professional development, and to revisit staffing and incentive policies for hardship areas.

Teacher training colleges and universities may find the results helpful in strengthening pre-service preparation, particularly in learner-centred teaching, competency-based assessment and teaching in multi-grade or poorly resourced classrooms. The County Government of Wajir could use the evidence when planning support for basic education and when making the case for more national resources. Closer to the classroom, head teachers and Curriculum Support Officers in Habaswein could use the findings to plan school-based professional learning, mentoring and supervision, while teachers may

gain a clearer sense of which parts of their practice need attention. For researchers, the study adds to a thin body of work on the CBC in marginalised settings and points to questions that deserve further inquiry.

1.9 Limitations of the Study

A few limitations are expected. Part of the data will come from teachers' own reports in questionnaires, and some teachers may rate their competence more highly than is warranted. To guard against this, the researcher will compare questionnaire responses with classroom observations and with interviews with head teachers and Curriculum Support Officers.

Access is another concern. Schools in the sub-county are far apart, roads are poor and some areas experience occasional insecurity. The researcher will plan fieldwork for the dry months, work closely with local education and security offices and engage research assistants who know the area well. Drought or the seasonal movement of pastoralist families may also close some schools for a time or reduce attendance. If that happens, a replacement school will be drawn from the same stratum. Lastly, because the study covers a single sub-county, its findings cannot simply be applied to the rest of Kenya, though they should speak to other ASAL areas with similar conditions.

1.10 Delimitations of the Study

The study is limited to public primary schools in Habaswein Sub-County, Wajir County. Private primary schools, pre-primary centres and junior schools are left out because they are staffed, funded and managed differently, which would make comparison difficult. Respondents will be TSC and Board of Management teachers who teach Grades 1 to 6, the head teachers of the sampled schools and the Curriculum Support Officers who serve the sub-county. The study focuses on four areas: teachers' knowledge and understanding of the CBC, their pedagogical competence, the factors that affect their competence, and the link between competence and effective implementation. Other influences, such as parental involvement and learner characteristics, are recognised but will not be examined in depth.

1.11 Assumptions of the Study

The study will be based on the following assumptions:

- i. That respondents will give honest and accurate information about their knowledge, practices and experiences in implementing the CBC.

- ii. That all public primary schools in Habaswein Sub-County are implementing the CBC in line with national guidelines issued by the Ministry of Education and KICD.
- iii. That teachers in the sampled schools have had at least some exposure to CBC training, whether through national rollout programmes, school-based sessions or self-directed learning.
- iv. That the security situation in the sub-county will remain stable enough to allow data collection within the planned period.

1.12 Operational Definition of Terms

Competency-Based Curriculum (CBC): The curriculum introduced in Kenya from 2017, which aims to develop learners' core competencies, values and practical skills through learner-centred teaching and continuous assessment. In this study, the term also covers the current reference to Competency-Based Education (CBE).

Teachers' competence: The mix of knowledge, teaching skills, assessment skills and attitudes that allows a teacher to plan, teach and assess CBC lessons as intended. It will be measured through teachers' knowledge of the CBC, their pedagogical competence and their assessment practice.

Knowledge and understanding of CBC: How well a teacher can explain the goals, core competencies, values, pertinent and contemporary issues, learning areas and assessment requirements of the CBC.

Pedagogical competence: A teacher's ability to plan and deliver lessons using learner-centred methods, build core competencies and values into lessons, manage learner activities and make good use of locally available resources.

Effective implementation of CBC: The extent to which classroom practice matches what the CBC intends, as shown by learner engagement, the integration of core competencies, the use of competency-based assessment, lesson planning in the prescribed format and timely coverage of learning areas.

Factors affecting competence: Conditions in the school, in the teacher and in the wider setting, such as training, workload, resources, supervision and the physical environment, that help or hinder a teacher in implementing the CBC.

Marginalised area: An area with poor access to social services, weak infrastructure and limited economic opportunity, as is the case in many ASAL counties, Wajir included.

1.13 Organisation of the Proposal

The proposal has three chapters. Chapter One introduces the study. It sets out the background, the problem, the purpose, the objectives, research questions and hypotheses, and then explains why the study matters, where its boundaries lie and what it assumes, before defining key terms. Chapter Two reviews the theories behind the study and the empirical literature related to each objective, and closes with the conceptual framework and the gaps the study will address. Chapter Three explains how the study will be done, covering the design, location, population, sampling, instruments, piloting, data collection and analysis, and ethical safeguards. References and appendices come at the end.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter looks at what other scholars have said about teacher competence and curriculum reform. It opens with three theories that help explain how teachers take up a new curriculum and what it means to be a competent teacher. A short description of Kenya's Competency-Based Curriculum follows. The chapter then reviews empirical studies under each of the four objectives, offers a critique of that body of work, presents the conceptual framework and ends by setting out the gaps this study hopes to fill.

2.2 Theoretical Review

The study draws on three theories. The Concerns-Based Adoption Model describes how teachers' worries and practices change as they get used to an innovation. The Teacher Professional Competence Model explains what competence is made of and how it turns into classroom performance. The Technological Pedagogical Content Knowledge framework describes the combined knowledge teachers need in order to teach particular content well with the tools at hand, a point that matters because digital literacy is one of the CBC's core competencies.

2.2.1 Concerns-Based Adoption Model

The Concerns-Based Adoption Model (CBAM) grew out of work by Gene Hall and his colleagues at the University of Texas in the 1970s. They wanted to understand how people, teachers in particular, experience and adopt new ideas in education. Their starting point was simple but important: change is a process, not an event, and it happens first in individuals before it becomes part of an institution. The model looks at change through three lenses, namely stages of concern, levels of use and innovation configurations (Okeyo & Mokua, 2023).

Stages of concern trace how a teacher's feelings about an innovation shift over time. Early on, teachers mostly want information. They want to know what the innovation is and how it will affect them. Once they start using it, their worries turn to practical matters such as time, materials and organisation. Later still, they begin to think about its effect on learners, about working with colleagues and about how to make it work better. Levels of use describe what teachers actually do, moving from non-use, through mechanical and routine use, to refinement and renewal. Innovation configurations capture the fact that different teachers often end up using the same innovation in quite different ways (Okeyo & Mokua, 2023).

The model has been used widely in studies of curriculum reform in Africa. In Kenya, Okeyo and Mokua (2023) applied it to teachers' preparedness for the CBC in Nyamira South and found that most teachers were ill-prepared for the new curriculum. For this study, CBAM offers a helpful way of reading teachers' knowledge and practice as points along a path of growth. A teacher with only a surface understanding of the CBC is likely to be stuck at an early, self-focused stage and to use the curriculum mechanically. A highly competent teacher, by contrast, is more likely to be thinking about learners' progress and adapting the curriculum with some creativity. The model also reminds us that teachers' concerns need the right kind of support, which ties in with the third objective on factors affecting competence.

CBAM is not without its critics. It concentrates on the individual teacher and says relatively little about structural barriers such as funding, class size, infrastructure and inconsistent policy, any of which can stall implementation no matter how willing a teacher is. It has also been faulted for implying a tidy, one-way progression, when in reality teachers can slip back to earlier concerns as conditions change. For these reasons, the study pairs CBAM with a model of teacher competence and pays close attention to the particular conditions in Habaswein.

2.2.2 Teacher Professional Competence Model

The second theory is the Teacher Professional Competence Model. It draws on the work of German and international researchers, Sigrid Blömeke and her colleagues among them, who argued that competence is best seen as a continuum. At one end are a teacher's dispositions: professional knowledge, but also beliefs, motivation and self-regulation. These feed into situation-specific skills, meaning the ability to notice what is happening in a lesson, make sense of it and decide what to do next. Those skills, in turn, show up as observable performance in the classroom (Jentsch & König, 2022).

Jentsch and König (2022) divide competence into two broad parts. The cognitive part covers content knowledge, pedagogical content knowledge and general pedagogical knowledge. The affective-motivational part covers teachers' beliefs about teaching and learning, their enthusiasm and their confidence in their own ability. Evidence from large international studies suggests that both parts affect the quality of teaching and what learners achieve, and that both can be developed during initial training and through ongoing professional learning. The lesson for this study is that knowledge alone is not enough. A teacher may know the principles of the CBC perfectly well and still struggle to apply them if confidence is low, if they doubt the curriculum's value or if they find it hard to read and respond to what is happening in class.

The model is useful here in three ways. It gives a clear structure to the independent variable, allowing competence to be split into knowledge and understanding of the CBC on the one hand and pedagogical competence on the other. It also explains why simply counting how many teachers attended training says little about competence, a weakness in much of the earlier CBC research. And because it treats competence as something that becomes visible in practice, it supports the decision to observe lessons as well as to use questionnaires. Its main weakness is that it was developed mostly in well-resourced school systems. It may therefore underplay how much hard conditions in marginalised areas can limit even very capable teachers, which is why the study examines contextual factors under the third objective.

2.2.3 Technological Pedagogical Content Knowledge Framework

The third lens is the Technological Pedagogical Content Knowledge (TPACK) framework, proposed by Punya Mishra and Matthew Koehler in the mid-2000s as an extension of the idea of pedagogical content knowledge. It holds that good teaching draws on three kinds of knowledge at once: knowledge of content (what to teach), knowledge of pedagogy (how to teach it) and knowledge of technology (the tools that can support learning). Where these overlap, for instance in pedagogical content knowledge or technological pedagogical knowledge, we find the specialised understanding that sets an expert teacher apart from someone who simply knows a subject or enjoys using gadgets (Mishra et al., 2023).

In revisiting the framework, Mishra et al. (2023) stress that teachers also need contextual knowledge, an understanding of the circumstances, limits and opportunities of the setting in which they teach. That point carries real weight in Habaswein, where electricity, devices and internet access cannot be taken for granted. Digital literacy is one of the CBC's core competencies, and teachers are expected to build it into every learning area. Yet Murithi and Yoo (2021) found that many Kenyan primary teachers had received only basic computer training and lacked the capacity to use technology meaningfully in CBC lessons. TPACK therefore helps the study to ask whether teachers can bring together content, pedagogy and whatever technology is available, including low-cost alternatives, to develop the intended competencies. Because its knowledge areas overlap and are hard to measure one by one, the study will use TPACK to interpret findings rather than as a measurement tool.

2.3 The Competency-Based Curriculum in Kenya

Kenya turned to a competency-based curriculum partly because the 8-4-4 system had become heavily focused on examinations and was not producing school leavers with the practical skills and values a changing economy needs (Akala, 2021). The Basic

Education Curriculum Framework set out a vision of nurturing every learner's potential. It named seven core competencies and a set of national values, and it listed pertinent and contemporary issues (PCIs), such as health, the environment, citizenship and life skills, to be woven across learning areas. Learners are meant to discover, create and apply knowledge through activities, while teachers guide the process rather than serving as the only source of information (Thukia, 2025).

Thukia (2025) reads the reform through Basil Bernstein's ideas on curriculum and pedagogy. In this reading, the CBC moves away from a tightly controlled, teacher-led model towards one in which learners have more say over the pace and order of their learning. The shift is demanding. Teachers must be secure enough to let go of some control while still steering learning with purpose. Where that confidence or skill is missing, schools may end up with a hybrid: new terminology on the surface, old routines underneath.

Assessment is one of the pillars of the reform. Under the KNEC (2021) framework, teachers carry out school-based assessment with a variety of tools and record learners' progress at four levels: exceeding expectations, meeting expectations, approaching expectations and below expectations. The PWPER (2023) acknowledged that inadequate teacher preparation, an overloaded timetable and a shortage of resources had held implementation back. It recommended fewer learning areas and a stronger push on teacher capacity building. These recommendations make clear that teacher competence is widely seen as decisive for the reform. They also show that the curriculum is still changing, so teachers will need to keep updating what they know.

2.4 Empirical Review

This section reviews empirical studies under each of the four objectives. Where relevant, evidence from other countries comes first and Kenyan studies follow, so that the issues can be seen at both regional and national level.

2.4.1 Teachers' Knowledge and Understanding of the Competency-Based Curriculum

It is hard to implement a curriculum well without understanding what it is trying to do. Across Africa, teachers' understanding of competence-based reforms appears to be uneven. In their review of studies from seven countries, Wawire et al. (2026) found a clear gap between policy and practice: the reforms call for a shift from covering content to developing competences, yet much classroom practice still reflects the older approach. Kasuga and Kalolo (2025) point to inadequate training and support as a central problem in Tanzania, noting that many teachers had not been properly prepared

for the new way of working. In Uganda, Kidega et al. (2024) likewise traced many of the difficulties teachers faced in Gulu City back to insufficient training on the new curriculum.

Kenyan evidence points the same way. Mwang'ombe (2021) looked at teachers' understanding and skills in relation to the CBC and found that understanding was often vague and uneven. In practice, this kind of vagueness can mean a teacher is able to list the core competencies but is unsure how to develop them in a particular lesson. Working in public pre-primary schools in Nairobi City County, Isaboke et al. (2021) found that 65.9% of teachers had received no CBC training at all. The authors linked this lack of training, along with poor facilities and large classes, to weak implementation. Their study targeted 450 pre-primary teachers, 225 ECD centre managers and 225 head teachers, and the findings are all the more striking because Nairobi is one of the best-resourced counties in Kenya.

More recent studies give a mixed picture. In Machakos County, Mutua and Waweru (2023) found that yearly in-service training was considered adequate, yet shortages of learning materials and a high pupil-teacher ratio still got in the way. Karua et al. (2025) examined teachers' awareness of the tenets of the CBC in Kinangop Sub-County and found that a majority (52.3%) were not competent in preparing schemes of work in line with the curriculum. Being aware of a curriculum, it seems, is not the same as knowing how to plan for it.

Mwangi et al. (2026) add a further twist. Although 95% of the teachers in their study had been trained, 65% held negative views of the curriculum. This suggests that training may have passed on information without building real understanding or a sense of ownership, which fits the argument of Jentsch and König (2022) that competence involves beliefs and motivation as well as knowledge. In junior secondary schools in Bomet County, Keter and Wabuke (2025) found that teacher preparedness was significantly associated with the level of implementation, and that gaps in preparation continued to hold back the quality of that implementation.

Very little of this research has been done in the ASAL counties. Hussein (2023) reported that teachers in northern Kenya had little exposure to CBC training and that poor infrastructure and scarce resources made proper grading difficult. Ongubo (2025) found that more than 60% of public primary teachers in Township Ward, Garissa, had not received comprehensive in-service training. Neither study, however, measured teachers' knowledge systematically or linked it statistically to implementation. The present study will do both in Habaswein.

2.4.2 Teachers' Pedagogical Competence in Implementing the CBC

Pedagogical competence is about what teachers actually do in class: how they plan, how they organise activities, how they draw learners in and how they use the resources available. The CBC asks teachers to use learner-centred approaches such as inquiry, cooperative learning, role play, projects, field visits and problem solving, and to build values and PCIs into their lessons. Wawire et al. (2026) found that across African countries using competence-based curricula, classroom practice remained largely teacher-centred. Learners had few chances to work together, investigate or create. The reasons are partly practical, such as large classes, a lack of materials and weak preparation, and partly a matter of habits formed over many years of examination-driven teaching.

In Kenya, the APHRC (2022) evaluation offers one of the closest looks at CBC teaching in practice. It examined lessons in schools and tried to identify what an effective CBC lesson looks like, with features such as clear learning outcomes, active learner involvement and deliberate attention to core competencies. The evidence shows that such lessons are possible in Kenyan public schools, but their quality varies a great deal, and many lessons meet the curriculum's expectations only in part. In Kirinyaga County, Mohamed et al. (2022) found that geography teachers' pedagogical preparedness for the CBC was low.

In Sabatia Sub-County, Vihiga County, Alwang'a and Were (2026) looked at the state of teachers' pedagogical knowledge and its connection to CBC implementation in public primary schools. Their work ties the quality of implementation to the strength of teachers' pedagogical knowledge and marks it out as a key ingredient of effective delivery. Karua et al. (2025) found that teachers in Kinangop had limited capacity to choose suitable teaching and learning approaches for CBC classes. Mutai et al. (2024), surveying 605 respondents in Moiben Sub-County, reported that teacher preparedness had a significant influence on how the CBC was implemented.

Assessment is part and parcel of teaching under the CBC. Macheso, Kati, and Wafula (2024) found that teachers in western Kenya relied on written tests far more than on portfolios, projects and anecdotal records, even though the KNEC (2021) framework calls for a wide range of tools. In a companion study of Grade 6 science and technology teachers in Bungoma, Busia and Trans-Nzoia, Macheso, Wafula, and Kati (2024) found that teachers' self-efficacy was linked to how far they implemented competency-based assessment. Put plainly, teachers who felt confident about assessment tended to carry it out more fully. This matters because it shows that pedagogical competence depends not

only on skill but also on teachers' belief in their own ability, just as the Teacher Professional Competence Model would predict.

Digital teaching is a further dimension. Murithi and Yoo (2021) surveyed 351 teachers. Most saw ICT as necessary for the CBC, but they had received only basic computer training and found it hard to use technology in their lessons. In Habaswein, where power and connectivity are limited, the challenge is probably greater still. We know little about how teachers in such places understand and teach the digital literacy competency, or whether they turn to low-cost alternatives.

2.4.3 Factors Affecting Teachers' Competence in Implementing the CBC

The literature points to several sets of factors that shape teachers' competence. They can be grouped under training and professional development, resources and infrastructure, workload and staffing, supervision and support, attitudes and beliefs, and the wider context.

Training is the factor mentioned most often. UNESCO and the International Task Force on Teachers for Education 2030 (2024) argue that professional development only improves teaching when it is continuous, grounded in practice and responsive to what teachers need. In Kenya, by contrast, early CBC training was widely described as short and rushed. Having found most teachers in Nyamira South ill-prepared, Okeyo and Mokua (2023) warned that training should not be carried out hurriedly as a public relations exercise. Ongubo (2025) found that the reach of in-service training in Township Ward, Garissa, was limited. In Nairobi County, Mogere (2022) likewise identified teacher training as an important determinant of how well the CBC was being implemented.

Resources and infrastructure form the second set. Cheruiyot (2024) identified shortages of learning materials, inadequate classrooms and weak technological infrastructure as major problems in junior schools, together with resistance to change among teachers, parents and learners. In northern Kenya, Hussein (2023) found that poor infrastructure and a lack of learning resources made proper assessment difficult. Mutua and Waweru (2023) reported that a shortage of teaching and learning materials held back implementation in Machakos. When resources are this thin, even skilled teachers may be unable to use the methods they know would work best.

Workload and staffing come next, because they decide how much time and energy teachers have for planning, assessment and reflection. The CBC calls for detailed lesson plans and continuous assessment records for every learner, which can quickly become overwhelming in a large class. Isaboke et al. (2021) listed large classes and

understaffing among the main barriers in Nairobi, and Mutua and Waweru (2023) drew attention to the high pupil-teacher ratio. The shortage of teachers is sharper still in ASAL counties. The County Government of Wajir (2023) recognises it as a serious and continuing problem. In practice, many schools lean on Board of Management teachers, some of whom may never have received formal CBC training.

Supervision and support make up a fourth set. Awiti (2023), studying primary schools in Nairobi City, examined a range of school-level factors affecting CBC implementation and underlined the importance of the support teachers receive within their schools. It is reasonable to expect that where head teachers observe lessons regularly, review schemes of work and organise peer learning, teachers will grow in confidence. Keter and Wabuke (2025) also stress the value of continuous support in strengthening teacher preparedness.

Attitudes and beliefs are a fifth. Mwangi et al. (2026) found that most teachers held negative views of the CBC despite having been trained, and Cheruiyot (2024) identified resistance to change among teachers as a barrier. The Teacher Professional Competence Model suggests that such beliefs colour how teachers read classroom situations and how they respond. A teacher who doubts the value of learner-centred methods may slip back into lecturing, especially when there is pressure to finish the syllabus.

The last set concerns the wider context, which is especially important in marginalised areas. Barasa et al. (2025) showed that in Uganda, schools serving low-income communities found the competence-based reform harder to implement than schools serving wealthier families, and they warned that without deliberate support the reform could lock in inequality. Kasuga and Kalolo (2025) made a similar point about rural schools in Tanzania. In northern Kenya, drought, pastoralist movement, long distances and insecurity disrupt both learners' attendance and teachers' access to training (County Government of Wajir, 2023; Hussein, 2023). Such conditions can wear down morale and push up staff turnover, breaking the continuity that competence needs in order to grow. Few studies have looked at how these conditions interact with teacher competence under the CBC in Kenya, and this study aims to do so.

2.4.4 Teachers' Competence and Effective Implementation of the CBC

The fourth objective concerns the link between teachers' competence and how effectively the CBC is implemented. A number of Kenyan studies have explored related links, though most used preparedness rather than competence as the explanatory variable. Mutai et al. (2024) found a significant positive relationship between teacher preparedness and CBC implementation in Moiben Sub-County. Keter and Wabuke (2025) reported that teacher preparedness significantly influenced implementation in

public junior secondary schools in Bomet County. Mutua and Waweru (2023) examined teachers' competence, in terms of values, attitudes, knowledge and skills, alongside the extent of CBC implementation in Machakos County. They concluded that implementation depended both on the teachers and on the conditions in which they worked, such as the supply of materials and the pupil-teacher ratio.

Alwang'a and Were (2026) narrowed the focus to pedagogical knowledge as one specific dimension of competence. Their work suggests that the state of teachers' pedagogical knowledge has a direct bearing on how the CBC is implemented in Sabatia, which strengthens the case for targeted professional development in learner-centred methods. Macheso, Wafula, and Kati (2024) found that teacher efficacy was linked to the implementation of competency-based assessment. The APHRC (2022) evaluation also suggests that the quality of CBC lessons depends heavily on what teachers bring into the classroom, including how they plan and how well they understand the intended competencies.

Beyond Kenya, the review by Wawire et al. (2026) shows that the gap between competence-based policy and classroom practice is closely bound up with teacher preparation and the resources available to schools. In Uganda, Kidega et al. (2024) recommended sustained training and support for teachers, noting that many of the challenges they faced came from gaps in their preparation for the new curriculum. Jentsch and König (2022) supply the theoretical thread that ties these findings together: teacher competence affects the quality of teaching, and the quality of teaching affects what learners gain.

Useful as this evidence is, it has three weaknesses. Most studies measured competence or preparedness through self-report questionnaires alone, which may paint too rosy a picture. Few separated the different dimensions of competence or tested what each contributes to implementation. And almost none were carried out in the ASAL counties of North Eastern Kenya. This study responds by combining questionnaires with classroom observation and interviews, by testing the separate and combined effects of knowledge and pedagogical competence, and by focusing on Habaswein Sub-County.

2.5 Critique of Existing Literature

Taken together, the literature agrees that teacher competence is central to the success of the CBC and that many Kenyan teachers have yet to acquire the full set of competences the curriculum requires. The body of work does, however, have its weak spots. The most obvious is geographical. Most studies have been carried out in Nairobi, central Kenya, the Rift Valley and western Kenya (Isaboke et al., 2021; Karua et al., 2025; Mohamed et al., 2022; Mutai et al., 2024). The handful of studies from northern Kenya,

such as Hussein (2023) and Ongubo (2025), were mainly descriptive and did not test how competence relates to implementation.

There is also some confusion over concepts. Preparedness, readiness, competence and capacity are often used as if they meant the same thing, and many studies gauged preparedness mainly by asking whether teachers had attended training. The work of Mwangi et al. (2026) shows how weak that measure is, since trained teachers may still lack understanding or hold negative views of the curriculum. This study uses a clearer definition of competence, drawn from the Teacher Professional Competence Model (Jentsch & König, 2022), which separates knowledge from pedagogical competence.

Method is a third concern. Many studies relied on questionnaires alone, which invite respondents to present themselves in a good light. Only a few, among them APHRC (2022) and Okeyo and Mokua (2023), used observation or document analysis to check what teachers said against what they did. Some studies also worked with small or purposively chosen samples, which limits how far their findings can be generalised. Mohamed et al. (2022), for example, drew their sample from a target population of just 24 schools. This study will use stratified random sampling, an adequate sample and several sources of data to make its findings more credible.

Lastly, context has usually been treated as background rather than as something that interacts with teacher competence. Barasa et al. (2025) and Kasuga and Kalolo (2025) show that context shapes how competence-based reforms play out, yet Kenyan studies have seldom asked how drought, insecurity and pastoralist movement affect teachers' professional growth. This study will look at these factors directly.

2.6 Conceptual Framework

Figure 2.1 shows the relationship the study expects to find between the independent, intervening and dependent variables.

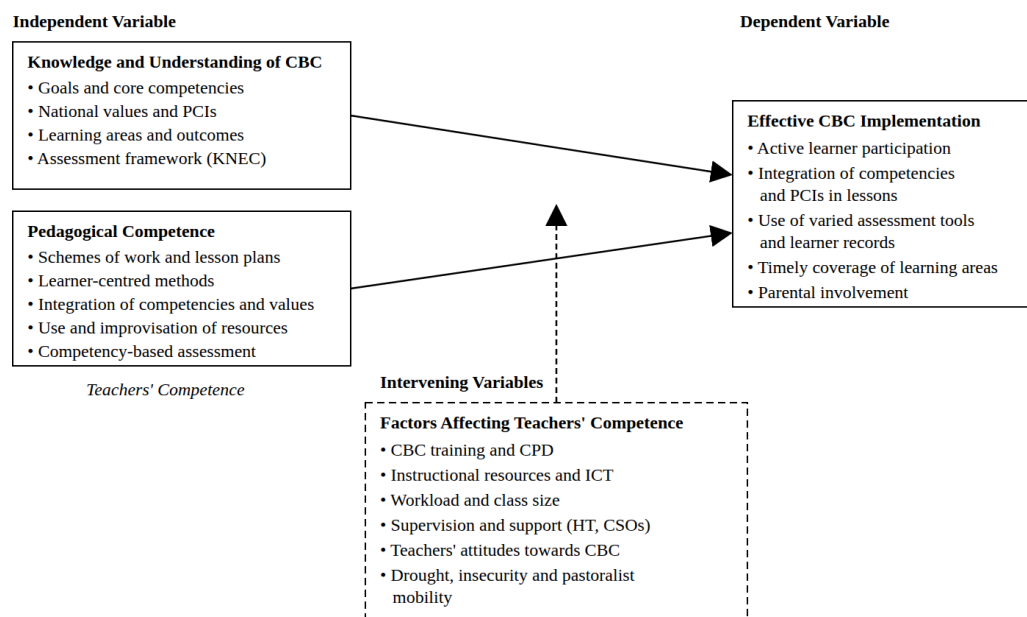


Figure 2.1: Conceptual Framework

Source: Researcher (2026)

The independent variable is teachers' competence, which has two dimensions. The first is knowledge and understanding of the CBC: how well teachers grasp the curriculum's goals, core competencies, values, PCIs, learning areas and assessment requirements. The second is pedagogical competence: teachers' ability to prepare schemes of work and lesson plans in the required format, teach through learner-centred methods, build core competencies and values into lessons, use and improvise resources and carry out competency-based assessment.

The dependent variable is the effective implementation of the CBC. It will be measured through indicators that reflect what the curriculum sets out to achieve: active learner participation, the integration of core competencies and PCIs in lessons, the use of varied assessment tools and proper learner records, coverage of learning areas on schedule and parents' involvement in learning.

The factors affecting teachers' competence are treated as intervening variables. They include how adequate CBC training and continuous professional development have been, the availability of teaching resources and ICT, teachers' workload and class size, supervision and support from head teachers and Curriculum Support Officers, teachers' attitudes towards the CBC and local conditions such as drought, insecurity and pastoralist movement. These factors can strengthen or weaken the link between competence and implementation. A knowledgeable teacher, for instance, may still find group work almost impossible with 90 learners and no desks. Government policy,

including the reforms recommended by the PWPER (2023), is acknowledged as a wider influence but will not be measured directly.

2.7 Summary of Literature and Research Gaps

The review shows that the CBC asks a great deal of teachers: a solid understanding of the curriculum, skill in learner-centred teaching and the ability to assess competences. Each of the three theories sheds light on a different part of this demand. CBAM shows that teachers take up new ideas gradually and need support that matches their concerns. The Teacher Professional Competence Model shows that competence is a blend of knowledge, beliefs and situation-specific skills. TPACK draws attention to the combined knowledge needed to teach content well with the tools available in a given setting. Studies from Kenya and elsewhere in Africa confirm that gaps in teacher competence are common and that training, resources, workload, supervision, attitudes and context all play a part. Table 2.1 summarises the main studies and the gaps they leave.

Table 2.1: Summary of Research Gaps

Author(s) and Year	Focus of the Study	Key Findings	Research Gap and How the Study Addresses It
Isaboke et al. (2021)	Teacher preparedness and CBC implementation in pre-primary schools, Nairobi	65.9% of teachers had not received CBC training; large classes and inadequate resources hindered implementation	Urban, pre-primary focus; current study targets primary schools in a marginalised ASAL setting
Okeyo and Mokua (2023)	Preparedness of teachers for CBC in secondary schools, Nyamira South	Schools were ill-prepared; most teachers were untrained; CBAM used	Secondary school focus; current study examines primary schools and measures competence directly
Mutua and Waweru (2023)	Teachers' competence and CBC implementation, Machakos County	Training considered adequate, but materials and high pupil-teacher ratio constrained implementation	Relatively well-served county; current study examines an ASAL context with different constraints
Hussein (2023)	Challenges of CBC implementation in northern Kenya	Poor infrastructure, few resources and limited training undermined implementation	Descriptive study; current study tests the statistical relationship between competence and implementation

Author(s) and Year	Focus of the Study	Key Findings	Research Gap and How the Study Addresses It
Macheso, Kati, and Wafula (2024)	Use of competency-based assessment tools in western Kenya	Teachers relied on a narrow range of assessment tools	Focused on assessment only; current study includes assessment as part of overall pedagogical competence
Karua et al. (2025)	Teacher preparedness for CBC, Kinangop Sub-County	52.3% of teachers were not competent in preparing schemes of work	Lower primary focus in a highland county; current study covers Grades 1 to 6 in Wajir
Ongubo (2025)	Teacher preparedness for CBC adoption, Garissa County	Over 60% of teachers lacked comprehensive in-service training	Limited to one ward; did not link competence to implementation outcomes
Mwangi et al. (2026)	Teacher training and perceptions prior to CBC implementation	95% trained but 65% held negative perceptions	Did not assess classroom practice; current study uses observation to verify competence
Wawire et al. (2026)	Scoping review of CBC pedagogy and assessment across Africa	Classroom practice remains largely teacher-centred	Regional synthesis; current study provides primary data from a specific marginalised context

Source: Researcher (2026)

Three gaps stand out. The first is contextual: no known study has examined teacher competence and CBC implementation in Habaswein Sub-County, or in Wajir County more broadly. The second is conceptual, since many studies have used training attendance as a stand-in for competence. The third is methodological, as few studies have checked self-reported data against classroom observation or tested what each dimension of competence contributes on its own. This study has been designed with all three gaps in mind.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains how the study will be carried out. It covers the research design and the setting, the people who will take part and how they will be chosen, and the instruments that will be used. It also describes how the instruments will be piloted and checked, how data will be gathered and analysed, the practical arrangements for fieldwork and the ethical safeguards that will guide the whole process.

3.2 Research Design

The study will use a convergent parallel mixed methods design that combines a descriptive survey with a correlational component. In this kind of design, quantitative and qualitative data are gathered during the same period and analysed separately. The two sets of results are then brought together at the interpretation stage, so that each can confirm, extend or question the other (Creswell & Creswell, 2023). The descriptive survey will show the current level of teachers' knowledge and pedagogical competence and the factors that affect them, as they are, without any attempt to manipulate variables. The correlational part will test how strongly, and in which direction, teachers' competence is related to effective CBC implementation, as set out in the hypotheses.

The design suits the study for a few reasons. The study needs both to describe a situation and to test relationships, and that calls for quantitative data from a fairly large number of teachers. Competence, however, is a complex idea, and questionnaires alone may not capture it faithfully. Interviews with head teachers and Curriculum Support Officers, together with classroom observation, will add depth and make the evidence more believable. The approach also has a track record in Kenyan research on the CBC. Okeyo and Mokua (2023) used a convergent mixed methods design to study teacher preparedness in Nyamira South, while Mutua and Waweru (2023) used a descriptive survey to study teacher competence in Machakos. The main demand of the design is that the two strands of data must be properly brought together. The researcher will handle this by preparing a joint display of the findings during analysis.

3.3 Location of the Study

The study will take place in Habaswein Sub-County, one of the sub-counties of Wajir County in Kenya's North Eastern region. It lies in the southern part of the county, along the main road between Garissa and Wajir towns, and shares a boundary with Garissa County to the south and west. The land is mostly dry. Rainfall is low and unpredictable,

temperatures are high and droughts come round often. Most residents are pastoralists who keep camels, cattle, goats and sheep, and many households move with the seasons in search of water and pasture. Habaswein town is the administrative and trading centre, while the rest of the population lives in small settlements scattered over a very large area (County Government of Wajir, 2023).

Wajir County has some of the weakest education indicators in the country. Its County Integrated Development Plan names low enrolment and transition rates, too few teachers, inadequate classrooms and sanitation facilities, and the disruption caused by drought and insecurity as key challenges (County Government of Wajir, 2023). Several public primary schools in the sub-county run low-cost boarding facilities to keep children from mobile households in school. Habaswein was chosen for three reasons. It typifies the marginalised ASAL setting in which the CBC is thought to be hardest to implement. No earlier study has examined teacher competence and CBC implementation there. And the researcher knows the area well, which should make it easier to reach schools and communicate with respondents.

3.4 Target Population

The target population is made up of all public primary schools in Habaswein Sub-County, their head teachers and teachers, and the Curriculum Support Officers (CSOs) who serve the sub-county. The school will be the unit of analysis, while teachers, head teachers and CSOs will be the units of observation. Teachers are included because they are the ones who deliver the CBC day to day. Head teachers lead teaching and learning in their schools, supervise staff and decide how resources are used. CSOs support teachers professionally and monitor how the curriculum is being implemented across schools.

Preliminary information gathered by the researcher suggests that the sub-county has about 42 public primary schools, an estimated 336 teachers (both TSC and Board of Management) and 3 CSOs. These figures are provisional and will be confirmed with the Sub-County Director of Education and the TSC Sub-County Director before data collection starts. Table 3.1 shows how the population is distributed.

Table 3.1: Target Population

Category of School (Stratum)	Number of Schools	Number of Head Teachers	Number of Teachers
Schools in Habaswein town and its environs	12	12	120
Schools in rural settlements (including low-cost boarding schools)	30	30	216

Category of School (Stratum)	Number of Schools	Number of Head Teachers	Number of Teachers
Total	42	42	336

Source: Provisional figures, to be confirmed with the Habaswein Sub-County Education Office (2026)

The 3 Curriculum Support Officers serving Habaswein Sub-County will also form part of the target population, in addition to the groups shown in Table 3.1.

3.5 Sampling Techniques and Sample Size

3.5.1 Sampling Techniques

Several sampling methods will be combined: stratified random sampling, simple random sampling, proportionate allocation and a census. Schools will first be split into two strata by location, namely schools in Habaswein town and its surroundings and schools in rural settlements. The split matters because schools in and near the town probably enjoy better transport, electricity, access to training and supervision than those further out, and such differences may shape teachers' competence. Within each stratum, schools will then be picked by simple random sampling, using a table of random numbers or a computer-generated list. This gives every school the same chance of selection and reduces bias (Creswell & Creswell, 2023).

Half of the schools in each stratum will be selected, which means 6 from the town stratum and 15 from the rural stratum, or 21 schools in all. Taking half of a fairly small number of schools gives good coverage of both strata while keeping the work within the time and money available. Because each school has only one head teacher, the head teachers of the 21 sampled schools will automatically form part of the sample. All 3 CSOs will be included through a census, as there are few of them and their view of the support available to teachers is important.

3.5.2 Sample Size

The number of teachers to be sampled will be worked out with the following formula, which is widely used in educational surveys when the size of the population is known:

$$n = N / (1 + N(e)^2)$$

Here, n is the required sample size, N is the target population of teachers (336) and e is the level of precision, set at 0.05 for a 95% confidence level. Putting in these values gives:

$$n = 336 / (1 + 336(0.05)^2) = 336 / 1.84 = 182.6 \approx 183 \text{ teachers}$$

The 183 teachers will be shared between the two strata in proportion to their size, using the formula: $\text{stratum sample} = (\text{stratum population} / \text{total population}) \times 183$. In each sampled school, teachers will be listed from the staff register and chosen by simple random sampling until the number needed for that stratum is reached. If a school has fewer teachers than required, all of them will be included and the shortfall made up from other sampled schools in the same stratum. Table 3.2 shows the resulting sample.

Table 3.2: Sample Size Distribution

Category of Respondents	Target Population	Sample Size	Sampling Technique
Teachers: Habaswein town and environs	120	65	Stratified, proportionate and simple random
Teachers: Rural settlements	216	118	Stratified, proportionate and simple random
Head teachers	42	21	Head teachers of the sampled schools
Curriculum Support Officers	3	3	Census
Total	381	207	

Source: Researcher (2026)

On top of this, the researcher will observe 21 CBC lessons, one in each sampled school, using a classroom observation checklist. In each school, the teacher to be observed will be picked at random from the sampled teachers who handle Grades 4 to 6, where most upper primary learning areas and school-based assessments take place. Taking part in the observation will be voluntary.

3.6 Research Instruments

Data will be collected with three instruments: a questionnaire for teachers, an interview guide for head teachers and CSOs, and a classroom observation checklist. Using more than one instrument will make it possible to cross-check the data and will add to the credibility of the findings.

3.6.1 Questionnaire for Teachers

The teachers' questionnaire (Appendix III) will be the main source of quantitative data. It has six sections. Section A asks about the respondent: gender, age, academic and professional qualifications, teaching experience, employer (TSC or Board of Management), grades taught and any CBC training attended. Section B deals with

knowledge and understanding of the CBC, and Section C with pedagogical competence, covering planning, learner-centred methods, the integration of competencies and assessment. Section D looks at the factors that affect teachers' competence, and Section E at how effectively the CBC is being implemented. Section F offers open-ended questions so that teachers can share their views in their own words.

Sections B to E use five-point Likert items that run from 1 (Strongly Disagree) to 5 (Strongly Agree). Section B also includes a short set of true-or-false knowledge items, so that knowledge is not measured by self-rating alone. A questionnaire makes sense for this study because it can reach many respondents in a short time, it protects their anonymity and it produces data that can be coded and analysed statistically without difficulty (Creswell & Creswell, 2023). It will be given in English, which is both the language of instruction and teachers' official working language.

3.6.2 Interview Guide for Head Teachers and Curriculum Support Officers

The 21 head teachers and 3 CSOs will take part in semi-structured interviews based on the guide in Appendix IV. The questions ask about teachers' knowledge and classroom practice, the support available to them, the difficulties of implementing the CBC and ideas for improvement. A semi-structured format lets the researcher follow up on interesting answers and ask for clarification while keeping the conversation on the study's objectives. Each interview should take 30 to 45 minutes. With the interviewee's permission, it will be audio-recorded and transcribed later; where permission is refused, the researcher will take detailed notes instead.

3.6.3 Classroom Observation Checklist

A structured checklist (Appendix V) will be used to record what actually happens in CBC lessons. It looks for evidence of lesson planning, learner-centred methods, the integration of core competencies, values and PCIs, the use of teaching and learning resources, learner participation and assessment. Each item is rated from 1 (Not observed) to 4 (Fully observed), and the observer will also keep short field notes. Observation offers a check on what teachers say in the questionnaire and gives direct evidence of implementation. To lessen the effect of having a visitor in the room, the researcher will arrive a few minutes before the lesson starts and sit quietly at the back without taking part.

3.6.4 Operationalisation of Study Variables

Table 3.3 sets out how each variable will be measured and analysed.

Table 3.3: Operationalisation of Study Variables

Variable	Indicators	Measurement Scale	Instrument	Analysis Technique
Knowledge and understanding of CBC (Independent)	Understanding of goals, core competencies, values, PCIs, learning areas and assessment framework	Ordinal (Likert); knowledge test score	Questionnaire ; interview guide	Frequencies, percentages, means, standard deviation; Pearson correlation; regression
Pedagogical competence (Independent)	Schemes of work and lesson plans; learner-centred methods; integration of competencies; use of resources; competency-based assessment	Ordinal (Likert); observation rating	Questionnaire ; observation checklist	Means, standard deviation; Pearson correlation; regression
Factors affecting competence (Intervening)	Training and CPD; resources and ICT; workload and class size; supervision; attitudes; drought, insecurity and mobility	Ordinal (Likert); open-ended responses	Questionnaire ; interview guide	Frequencies, percentages, means; thematic analysis
Effective CBC implementation (Dependent)	Learner participation; integration of competencies and PCIs; use of varied assessment tools and records; coverage of learning areas; parental involvement	Ordinal (Likert); observation rating	Questionnaire ; observation checklist; interview guide	Means, standard deviation; regression; thematic analysis

Source: Researcher (2026)

3.7 Pilot Study

The instruments will be tried out first in neighbouring Wajir South Sub-County, which is similar to Habaswein in climate, way of life and schooling but falls outside the main study. The pilot will involve roughly 10% of the sample: 18 teachers, 2 head teachers and 2 classroom observations. It will help the researcher spot questions that are unclear, repeated or sensitive, find out how long each instrument takes, check that the instructions make sense and test the procedures for data entry and coding. The instruments will then be revised in light of what the pilot shows, before the main fieldwork begins.

3.7.1 Validity of the Research Instruments

Validity is about whether an instrument measures what it is meant to measure. This study is concerned mainly with content validity and construct validity. For content validity, the instruments will be reviewed by the researcher's supervisors and by two

curriculum studies specialists at Mount Kenya University. Each reviewer will rate how relevant every item is to the study objectives on a four-point scale. A Content Validity Index (CVI) will then be worked out by dividing the number of items rated relevant or highly relevant by the total number of items. An index of 0.78 or more will be taken as acceptable, and items that fall below this mark will be reworded or dropped. Construct validity will be supported by drawing questionnaire items from indicators found in the theories and studies reviewed in Chapter Two, and by checking in the pilot that items meant to measure the same idea are correlated.

3.7.2 Reliability of the Research Instruments

Reliability is about consistency: whether an instrument gives similar results under similar conditions. The internal consistency of the Likert sections of the teachers' questionnaire will be checked with Cronbach's alpha, calculated from the pilot data. A coefficient of 0.70 or higher will be regarded as acceptable, in line with common practice in social science research (Field, 2024). If a section falls short, its item-total statistics will be examined and weak items revised or removed. To make the observation checklist more reliable, the researcher and a trained research assistant will observe the same pilot lessons separately and compare their ratings. Where they differ, they will talk through the differences and sharpen the checklist descriptors until they agree closely.

3.7.3 Trustworthiness of Qualitative Data

The qualitative data will be judged against four criteria: credibility, dependability, confirmability and transferability. Credibility will come from comparing what teachers, head teachers, CSOs and classroom observations reveal, and from member checking, in which some interviewees will be shown a summary of their interview and asked whether it reflects their views accurately. Dependability will be supported by an audit trail recording the decisions made during data collection and analysis. For confirmability, the researcher will keep reflective notes on personal assumptions that might colour the interpretation. Transferability will be helped by a rich description of the study setting, so that readers can decide for themselves whether the findings apply elsewhere (Braun & Clarke, 2022).

3.8 Data Collection Procedures

The first step will be to obtain an introduction letter from the School of Education at Mount Kenya University and ethical approval from the University's Ethics Review Committee. The researcher will then apply for a research permit from the National Commission for Science, Technology and Innovation (NACOSTI). With the permit in

hand, the researcher will report to the Wajir County Commissioner and the County Director of Education for authority to work in the county, and then to the Sub-County Director of Education and the TSC Sub-County Director in Habaswein. Head teachers of the sampled schools will be contacted beforehand to explain the study and to agree on suitable dates for visits.

Two research assistants will be recruited. They will be qualified teachers who know the area and speak both English and Somali, and they will receive two days of training on the study objectives, the instruments, ethical procedures and personal safety. At each school, the team will first meet the head teacher, explain the study and seek informed consent from participants. Questionnaires will be handed to the sampled teachers and, wherever possible, collected the same day to keep the response rate high. Teachers who ask for more time will have up to three days. Head teachers will be interviewed in a private room at the school and CSOs at their offices. Lesson observations will take place at times agreed with the teachers. Fieldwork should take about six weeks. It will be timed to fall within the school term, away from examination periods and, as far as possible, away from the height of the dry season, when some families move away with their animals.

3.9 Data Analysis and Presentation

3.9.1 Quantitative Data Analysis

Once fieldwork is over, the completed questionnaires and observation checklists will be checked for completeness and consistency, coded and entered into the Statistical Package for the Social Sciences (SPSS), version 29. The data will then be cleaned by looking for missing values, outliers and entry errors. Frequencies, percentages, means and standard deviations will be used to describe the respondents and to answer the first three research questions. Mean scores on the Likert items will be read as follows: 1.00 to 1.80 very low, 1.81 to 2.60 low, 2.61 to 3.40 moderate, 3.41 to 4.20 high and 4.21 to 5.00 very high.

The hypotheses under the fourth objective will be tested with inferential statistics. Pearson's product-moment correlation will show how strong the relationship is, and in which direction, between each dimension of teachers' competence and effective CBC implementation (H01 and H02). Multiple linear regression will then show the combined and separate influence of the two dimensions of competence on effective implementation (H03). The regression model is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

In this model, Y is effective CBC implementation, β_0 is the constant, β_1 and β_2 are the regression coefficients, X_1 is teachers' knowledge and understanding of the CBC, X_2 is teachers' pedagogical competence and ϵ is the error term. Each variable's composite score will be the mean of its items. Before running the regression, the researcher will check that its assumptions hold, namely normality, linearity, homoscedasticity, independence of errors and the absence of multicollinearity, using histograms, scatter plots, the Durbin-Watson statistic and variance inflation factors (Field, 2024). The coefficient of determination (R^2) will show how much of the variation in implementation is explained by teachers' competence, and the F-test will show whether the model as a whole is significant. All hypotheses will be tested at the 0.05 level of significance. Results will be presented in tables, bar charts and pie charts, each followed by interpretation and discussion.

3.9.2 Qualitative Data Analysis

Data from interviews, open-ended questionnaire items and observation notes will be analysed through reflexive thematic analysis, following the six phases set out by Braun and Clarke (2022). These are getting familiar with the data, generating initial codes, building themes, reviewing them, defining and naming them, and writing up. Interview recordings will be transcribed word for word. Any answers given in Somali will be translated into English, and a second translator will check the translation. The researcher will read the transcripts several times, code them and group the codes into themes linked to the study objectives, using NVivo to keep the codes and themes organised. Findings will be reported as a narrative supported by direct quotations. Participants will be identified only by codes, for example HT1 for the first head teacher and CSO2 for the second Curriculum Support Officer.

3.9.3 Integration of Quantitative and Qualitative Findings

As the convergent design requires, the quantitative and qualitative results will be brought together at the interpretation stage. For each objective, the researcher will prepare a joint display that sets the main statistical results beside the related qualitative themes and observation evidence. Where the two agree, confidence in the findings grows. Where they differ, for example if teachers rate their competence higher than what is seen in their lessons, the researcher will look into the reasons and explain them (Creswell & Creswell, 2023).

3.10 Logistical Considerations

The long distances and security situation in the sub-county mean that fieldwork will need careful planning. The researcher will hire a dependable vehicle and plan visits so

that schools close to one another are covered on the same day. Travel will take place only in daylight, and the local administration and security officers will be consulted before visits to remote schools. The research assistants will come from the local community, which should make communication easier and help build trust. The work plan and budget appear in Appendices VI and VII.

3.11 Ethical Considerations

The study will follow the ethical standards expected of research with human participants. Ethical approval will be sought from the Mount Kenya University Ethics Review Committee, and a NACOSTI research permit will be obtained before any data are collected. Permission will also be sought from the county and sub-county education authorities and from the head teachers of the sampled schools.

Taking part will be entirely voluntary. Every participant will receive an information sheet and consent form (Appendix II) explaining what the study is about, what taking part involves, how long it will take and that they may withdraw at any time without penalty. Written consent will be obtained before any questionnaire is completed, any interview is held or any lesson is observed. Teachers will be assured that the observation is not an appraisal and that nothing seen will be passed to their employer or used for disciplinary purposes.

Confidentiality and anonymity will be protected throughout. Respondents will not write their names on the questionnaires, and schools and individuals will be referred to by codes in every report. Recordings, transcripts and completed instruments will be kept securely, and electronic files will be password-protected in line with the Data Protection Act of 2019. The data will be used only for academic purposes and destroyed once the period required by the university has passed. The researcher will respect the cultural and religious norms of the community, including dress codes, prayer times and gender norms, and will hold interviews at times and places that suit participants. Lastly, all sources used in the study will be properly acknowledged using APA style, and the final report will be checked for plagiarism in line with university policy.

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APPENDICES

The following appendices contain the research instruments and supporting documents for the study.

APPENDIX I: LETTER OF INTRODUCTION

[Full Name of Student]

Mount Kenya University

School of Education

P.O. Box 342-01000

Thika

[Date]

The Head Teacher,

_____ Primary School,

Habaswein Sub-County,

Wajir County.

Dear Sir/Madam,

RE: REQUEST TO COLLECT RESEARCH DATA IN YOUR SCHOOL

I am a postgraduate student at Mount Kenya University pursuing a Master of Education degree in Curriculum Studies. As part of the requirements for the award of the degree, I am carrying out a study titled “Teachers’ Competence and Its Influence on the Effective Implementation of the Competency-Based Curriculum in Public Primary Schools in Habaswein Sub-County, Wajir County, Kenya.”

Your school has been selected to take part in the study. I kindly request your permission to administer questionnaires to some of your teachers, to interview you and to observe one CBC lesson at a time convenient to the school. Participation is voluntary, and the information collected will be treated with strict confidentiality and used for academic purposes only. Neither your name nor the name of your school will appear in any report.

Attached are copies of my introduction letter from the university and the research permit from NACOSTI. Thank you in advance for your cooperation.

Yours faithfully,

[Full Name of Student]

[Registration Number]

APPENDIX II: INFORMED CONSENT FORM

Study title: Teachers' Competence and Its Influence on the Effective Implementation of the Competency-Based Curriculum in Public Primary Schools in Habaswein Sub-County, Wajir County, Kenya.

Purpose of the study: The study seeks to assess teachers' competence in implementing the CBC and to establish how this competence influences the effectiveness of implementation. The findings are expected to inform teacher support programmes and education policy for marginalised areas.

What participation involves: If you agree to take part, you will be asked to complete a questionnaire (about 30 minutes), take part in an interview (about 40 minutes) or allow the researcher to observe one of your lessons, depending on your category of respondent.

Risks and benefits: There are no known risks in taking part in this study. The lesson observation is not an appraisal and will not be shared with your employer. Although you will not receive any payment, your views will contribute to improving teacher support in Habaswein and similar areas.

Confidentiality: Your name will not be recorded on any instrument. All information will be coded, stored securely and used only for this study.

Voluntary participation: Your participation is entirely voluntary. You may decline to answer any question or withdraw at any time without any penalty.

Contact: If you have questions about the study, please contact the researcher on [phone number] or [email address], or the Mount Kenya University Ethics Review Committee.

Declaration by participant

I have read (or have had read to me) the information above. I understand the purpose of the study and what my participation involves, and I voluntarily agree to take part.

Participant's signature: _____ Date: _____

Researcher's signature: _____ Date: _____

APPENDIX III: QUESTIONNAIRE FOR TEACHERS

Instructions

This questionnaire seeks information on teachers' competence and the implementation of the Competency-Based Curriculum in public primary schools in Habaswein Sub-County. Please answer all questions honestly. Do not write your name or the name of your school anywhere on this questionnaire. Tick (✓) the appropriate box or fill in the spaces provided.

Section A: Demographic Information

1. Gender: Male ☐ Female ☐
2. Age bracket (years): Below 30 ☐ 30–39 ☐ 40–49 ☐ 50 and above ☐
3. Highest professional qualification: P1 Certificate ☐ Diploma ☐ Bachelor's degree ☐ Master's degree ☐ Other (specify) _____
4. Employer: TSC ☐ Board of Management ☐ Other (specify) _____
5. Teaching experience (years): Below 5 ☐ 5–10 ☐ 11–15 ☐ Above 15 ☐
6. Grade(s) currently taught: Grade 1–3 ☐ Grade 4–6 ☐ Both ☐
7. Have you attended any CBC training? Yes ☐ No ☐
8. If yes, how many days of CBC training have you attended in total? 1–5 days ☐ 6–10 days ☐ More than 10 days ☐
9. Location of your school: Habaswein town and environs ☐ Rural settlement ☐

Section B: Teachers' Knowledge and Understanding of the CBC

Part I: Indicate the extent to which you agree with each statement. Key: SD = Strongly Disagree (1); D = Disagree (2); N = Neutral (3); A = Agree (4); SA = Strongly Agree (5).

No.	Statement	SD	D	N	A	SA
B1	I clearly understand the vision and mission of the Competency-Based Curriculum.					
B2	I can explain each of the seven core competencies of the CBC.					
B3	I understand how national values should be integrated into lessons.					
B4	I can identify pertinent and contemporary issues (PCIs) relevant to the learning areas I teach.					

No.	Statement	SD	D	N	A	SA
B5	I understand the learning areas and their specific learning outcomes for the grades I teach.					
B6	I understand the KNEC competency-based assessment framework and the four performance levels.					
B7	I understand the role of parents in supporting learning under the CBC.					
B8	I can interpret the curriculum designs for the learning areas I teach without difficulty.					

Part II: For each statement, tick True or False.

No.	Statement	True	False
B9	Under the CBC, the teacher's main role is to act as a facilitator of learning.		
B10	Learner performance in CBC is reported using four levels of achievement.		
B11	Core competencies should only be taught in specific learning areas such as Science and Technology.		
B12	Portfolios, rubrics and observation schedules are examples of CBC assessment tools.		
B13	Summative examinations are the only valid way to assess learners under the CBC.		

Section C: Teachers' Pedagogical Competence

Indicate the extent to which you agree with each statement. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
C1	I prepare schemes of work in the format required for the CBC.					
C2	I prepare lesson plans that clearly state specific learning outcomes and key inquiry questions.					
C3	I regularly use group work, discussion and cooperative learning in my lessons.					
C4	I use inquiry, projects and problem-solving activities to help learners discover knowledge.					
C5	I deliberately integrate core competencies and values into my lessons.					
C6	I improvise teaching and learning resources from locally available materials.					

No.	Statement	SD	D	N	A	SA
C7	I adapt my teaching to meet the needs of learners with different abilities.					
C8	I use a variety of assessment tools such as observation, checklists, rubrics and portfolios.					
C9	I keep up-to-date assessment records for every learner in my class.					
C10	I give learners feedback that helps them to improve their learning.					
C11	I integrate digital or low-cost technology into my lessons where possible.					

Section D: Factors Affecting Teachers' Competence in Implementing the CBC

Indicate the extent to which you agree that each of the following affects your competence in implementing the CBC. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
D1	The CBC training I received was adequate to prepare me to implement the curriculum.					
D2	I have regular opportunities for continuous professional development on the CBC.					
D3	My school has adequate textbooks, curriculum designs and teaching materials.					
D4	My school has adequate ICT facilities, electricity and internet to support CBC teaching.					
D5	My workload allows me enough time to plan and assess learners as required by the CBC.					
D6	The number of learners in my class allows me to use learner-centred methods effectively.					
D7	The head teacher regularly supervises and supports my CBC teaching.					
D8	Curriculum Support Officers visit and support teachers in my school.					
D9	I have a positive attitude towards the CBC.					
D10	Drought and the movement of pastoralist families disrupt my teaching and assessment.					
D11	Insecurity in the area affects my ability to attend training and deliver the curriculum.					
D12	Parents in this community support their children's learning activities under the CBC.					

Section E: Effective Implementation of the CBC

Indicate the extent to which you agree with each statement about CBC implementation in your class. Key: SD = 1; D = 2; N = 3; A = 4; SA = 5.

No.	Statement	SD	D	N	A	SA
E1	Learners actively participate in activities during my lessons.					
E2	Learners in my class demonstrate core competencies such as communication and collaboration.					
E3	PCIs and values are integrated in most of my lessons.					
E4	I use a variety of formative assessment tools as required by the CBC.					
E5	I cover the learning areas in the time allocated in the curriculum designs.					
E6	Learners complete projects and practical activities as required.					
E7	Parents participate in their children's learning activities when asked.					
E8	Overall, the CBC is being implemented effectively in my class.					

Section F: General Comments

1. In your opinion, what are the main challenges you face in implementing the CBC in this sub-county?

2. What kind of support would most improve your competence in implementing the CBC?

3. Any other comments on CBC implementation in Habaswein Sub-County?

Thank you for your time and cooperation.

APPENDIX IV: INTERVIEW GUIDE FOR HEAD TEACHERS AND CURRICULUM SUPPORT OFFICERS

Introduction: Thank you for agreeing to take part in this interview. The purpose is to understand your views on teachers' competence and the implementation of the CBC in Habaswein Sub-County. The interview will take about 40 minutes. With your permission, it will be recorded. Your responses will remain confidential.

Part A: Background

1. How long have you served as a head teacher or Curriculum Support Officer, and how long have you been in Habaswein Sub-County?
2. What CBC training have you personally received?

Part B: Teachers' Knowledge and Understanding of the CBC

3. How would you describe the level of understanding of the CBC among teachers in your school or zone?
4. Which aspects of the curriculum do teachers seem to understand well, and which do they find most difficult?

Part C: Teachers' Pedagogical Competence

5. How do teachers in your school or zone plan and deliver CBC lessons? What methods do they commonly use?
6. How well do teachers carry out competency-based assessment and keep learner records?
7. Can you give an example of a lesson you observed that reflected the CBC very well, or one that did not?

Part D: Factors Affecting Teachers' Competence

8. What training and professional development opportunities are available to teachers here? How adequate are they?
9. How do resources, staffing and class sizes affect teachers' ability to implement the CBC?
10. In what ways do local conditions such as drought, pastoralist mobility and insecurity affect teaching and learning?
11. What support do you provide to teachers, and what challenges do you face in providing it?

Part E: Competence and Effective Implementation

12. In your experience, how does the competence of a teacher affect the way the CBC is implemented in the classroom?
13. What differences do you notice between schools or classes where the CBC is implemented well and those where it is not?
14. What should the Ministry of Education, TSC, KICD and the County Government do to strengthen teachers' competence in this sub-county?

Thank you for your time and valuable contribution.

APPENDIX V: CLASSROOM OBSERVATION CHECKLIST

School code: _____ Grade: _____ Learning area: _____ Date: _____

Number of learners present: _____ Duration of lesson: _____

Rating key: 1 = Not observed; 2 = Partly observed; 3 = Largely observed; 4 = Fully observed

No.	Aspect Observed	1	2	3	4
1	Teacher has a lesson plan in the CBC format with clear learning outcomes				
2	Lesson begins with a key inquiry question or activity that connects to learners' experience				
3	Learners work in pairs or groups during the lesson				
4	Learners are engaged in inquiry, problem solving or hands-on activities				
5	Teacher acts as a facilitator rather than the main source of information				
6	At least one core competency is deliberately developed				
7	Values and/or PCIs are integrated into the lesson				
8	Teacher uses relevant teaching and learning resources, including improvised materials				
9	Technology (digital or low-cost) is used appropriately				
10	Teacher uses formative assessment during the lesson (questions, observation, checklists)				
11	Learners receive feedback on their work				
12	Learners with different abilities are supported				
13	Assessment records and learner portfolios are available and up to date				
14	Lesson concludes with a reflection or summary involving learners				

Observer's field notes:

APPENDIX VI: WORK PLAN

Table A.1: Work Plan, October 2026 – September 2027

Activity	Oct– Nov 2026	Dec 2026– Jan 2027	Feb– Mar 2027	Apr– May 2027	Jun– Jul 2027	Aug– Sep 2027
Proposal correction and defence	✓					
Ethical approval and NACOSTI permit		✓				
Pilot study and revision of instruments		✓				
Main data collection			✓			
Data entry, cleaning and analysis				✓		
Writing of Chapters Four and Five				✓	✓	
Submission of draft thesis to supervisors					✓	
Corrections, final defence and submission						✓

APPENDIX VII: RESEARCH BUDGET

Table A.2: Research Budget

Item	Quantity	Unit Cost (KES)	Total Cost (KES)
Stationery, printing and binding of proposal	6 copies	1,500	9,000
Printing of questionnaires and other instruments	250 sets	60	15,000
NACOSTI research permit	1	1,000	1,000
Training of research assistants	2 persons × 2 days	2,500	10,000
Research assistants' allowance (fieldwork)	2 persons × 30 days	1,500	90,000
Vehicle hire and fuel for fieldwork	30 days	6,000	180,000
Accommodation and meals for researcher	30 days	3,000	90,000
Pilot study expenses	Lump sum	20,000	20,000
Airtime and internet	Lump sum	8,000	8,000
Transcription and translation of interviews	24 interviews	1,000	24,000
Data analysis software and services	Lump sum	25,000	25,000
Printing and binding of final thesis	6 copies	2,000	12,000
Subtotal			484,000
Contingency (10%)			48,400
Grand Total			532,400